GreenFuel Intelligence System (GFIS): AI-Driven Biogas Infrastructure Planning & Optimization Framework

Organization: Chatake Innoworks Pvt. Ltd. **Division:** GreenWorks Division – Sustainable Energy Innovation Unit **Address:** Chatake Innoworks Pvt. Ltd., Neharu Industrial Estate, Solapur, Maharashtra, India. **Website:** https://www.chatakeinnoworks.com **Email:** admin@chatakeinnoworks.com **LinkedIn:** https://www.linkedin.com/company/chatake-innoworks-pvt-ltd
(Insert company logo placeholder)

1. Executive Summary

1.1 Contextual Overview: The Imperative for Renewable Intelligence

The global energy landscape is undergoing a tectonic shift, driven by the dual imperatives of decarbonization and energy security. For India, a nation that imports over 85% of its crude oil and 50% of its natural gas, the transition to domestic renewable sources is not merely an environmental choice but a strategic economic necessity. In this context, the anaerobic digestion (AD) sector—colloquially known as the biogas industry—occupies a unique intersection between waste management, renewable energy generation, and rural economic empowerment. With an estimated potential to generate 62 million tonnes of Compressed Biogas (CBG) annually from agricultural residue, cattle dung, and municipal solid waste, the sector promises to replace a significant fraction of imported fossil fuels while addressing the critical challenge of waste disposal.
However, the realization of this potential faces a critical bottleneck: operational inefficiency. Despite ambitious government frameworks like the Sustainable Alternative Towards Affordable Transportation (SATAT) initiative and the Galvanizing Organic Bio-Agro Resources Dhan (GOBARdhan) scheme, the ground reality reflects a landscape of underperformance. A substantial number of installed biogas plants, particularly in decentralized rural settings, suffer from "functional obsolescence"—they exist physically but operate at suboptimal yields or suffer catastrophic biological failures due to a lack of real-time process visibility and control.

1.2 The Innovation: GreenFuel Intelligence System (GFIS)

The GreenFuel Intelligence System (GFIS) is proposed as a comprehensive technological intervention designed to bridge this efficiency gap. Developed by the GreenWorks Division of Chatake Innoworks Pvt. Ltd., GFIS is an AI-Driven Biogas Infrastructure Planning & Optimization Framework. It transforms the traditional, passive biogas digester into a "smart," cyber-physical system capable of self-regulation and predictive logic.
At its core, GFIS integrates three technological pillars:

1. **Internet of Things (IoT) Perception Layer:** A high-fidelity sensor network that continuously monitors critical biochemical parameters—pH, temperature,

oxidation-reduction potential (ORP), and gas composition—providing a real-time digital pulse of the digester's health.

2. **Computer Vision Feedstock Analysis:** An edge-computing module utilizing Convolutional Neural Networks (CNNs) to classify and grade incoming organic waste, ensuring optimal Carbon-to-Nitrogen (C:N) ratios and preventing the ingress of inorganic contaminants that degrade system performance.
3. **Predictive Machine Learning Engine:** A cloud-based analytics suite employing algorithms like XGBoost and Long Short-Term Memory (LSTM) networks to forecast biogas yield and predict process failures (such as acidosis) days in advance, allowing for proactive rather than reactive maintenance.

1.3 Strategic Alignment and Feasibility

This proposal is meticulously structured to align with the strategic objectives of key stakeholders in the Indian ecosystem:

- **Government Compliance (SATAT & GOBARdhan):** GFIS acts as a digital enabler for these schemes by ensuring the operational reliability required to secure bank financing and meet the stringent Letters of Intent (LoI) requirements for CBG offtake. It supports the GOBARdhan objective of "Waste to Wealth" by verifying the quality of Fermented Organic Manure (FOM), unlocking secondary revenue streams.
- **Startup India Recognition:** The project meets the innovation, scalability, and employment generation criteria set by the Department for Promotion of Industry and Internal Trade (DPIIT), positioning Chatake Innoworks for tax exemptions (Section 80-IAC) and public procurement advantages.
- **Academic & Technical Excellence (DIPEX 2025):** Tailored for the DIPEX state-level exhibition, the project addresses the themes of "Computational Intelligence," "Energy," and "Waste Management." The methodology strictly adheres to the DIPEX project report guidelines, focusing on solution innovativeness, technical feasibility, and societal impact.

1.4 Commercial Value Proposition

The implementation of GFIS offers a compelling Return on Investment (ROI). By stabilizing the biological process, GFIS targets a 15-20% increase in gas yield and a 30% reduction in operational downtime. Furthermore, the system's ability to generate verified data trails facilitates the trading of carbon credits and simplifies the subsidy claim process for Central Financial Assistance (CFA), creating a robust financial ecosystem for plant owners.
In summary, GFIS is not just a monitoring tool; it is a critical infrastructure layer necessary to professionalize and scale India's bio-energy sector, turning volatile waste streams into a stable, predictable energy asset.

2. Problem Statement

2.1 The "Biogas Efficiency Paradox" in India

India stands at the precipice of a bio-energy revolution, yet the sector is plagued by a paradox: high resource availability coupled with low asset utilization. While the theoretical potential for biogas production is immense—estimated at 62 million tonnes of CBG per annum—the actual

production is a fraction of this capacity. The primary reason for this discrepancy is the high rate of failure and underperformance in small-to-medium scale biogas plants.
Research indicates that a significant percentage of biogas plants installed in rural India become non-functional shortly after commissioning. This failure is rarely due to the fundamental anaerobic digestion technology, which is mature, but rather due to operational mismanagement. The "Deenabandhu" and floating-dome models widely used in India are robust but passive; they lack the feedback loops necessary to handle the dynamic nature of biological waste treatment.

2.2 Critical Operational Challenges

2.2.1 Biological Instability and Lack of Visibility

Anaerobic digestion (AD) is a complex, multi-stage biochemical process involving the synergistic action of hydrolytic, acidogenic, acetogenic, and methanogenic bacteria. These microbial communities are extremely sensitive to environmental conditions.

- **The "Black Box" Problem:** In the current scenario, operators treat the digester as a "black box"—waste goes in, and gas comes out. There is little to no monitoring of internal parameters like pH, temperature, or Volatile Fatty Acids (VFA).
- **Methanogenesis Crash:** Methanogens, the bacteria responsible for methane production, are slow-growing and highly susceptible to pH drops (acidosis). Without sensors, operators cannot detect the onset of acidosis until gas production stops. By then, the bacterial colony is often dead, requiring a complete, costly, and time-consuming restart of the plant.
- **Temperature Sensitivity:** Rural plants often lack active temperature control. In winter, digester temperatures can drop below the mesophilic range (35°C-37°C), causing bacterial dormancy and a sharp decline in yield. The lack of predictive data means operators fail to take preemptive measures like insulation or heating.

2.2.2 Feedstock Variability and Contamination

The "Garbage In, Garbage Out" principle applies strictly to biogas.

- **Inorganic Contamination:** In community plants sourcing municipal waste, poor segregation leads to the ingress of plastics, glass, and metals. These inorganics accumulate in the digester, reducing the effective volume (hydraulic retention time) and causing mechanical failures in pumps and agitators. Manual segregation is labor-intensive and error-prone.
- **Nutrient Imbalance:** Efficient digestion requires a Carbon-to-Nitrogen (C:N) ratio of approximately 25-30:1. Different feedstocks have vastly different profiles (e.g., sawdust is high in Carbon, poultry litter is high in Nitrogen). Without real-time analysis of the input stream, operators cannot adjust the mixing ratio, leading to either ammonia toxicity (high N) or poor gas production (high C).

2.2.3 Economic Viability and Offtake Uncertainty

Despite the SATAT scheme guaranteeing offtake, the economics of CBG production are tight.

- **Pricing Pressure:** The procurement price of CBG is often benchmarked against CNG. Recent revisions have set the price around Rs. 1478/mmBtu or competitive per kg rates, but profitability hinges on maximizing yield. Any dip in efficiency directly erodes the thin

profit margins.

- **Maintenance Costs:** The industry relies on "breakdown maintenance"—fixing equipment only after failure. This is significantly more expensive than "predictive maintenance," which can foresee pump failures or blockages before they halt production.
- **Lack of Data for Financing:** Banks view biogas projects as high-risk due to the historical failure rate. The absence of credible, historical performance data makes it difficult for entrepreneurs to secure loans, even under the "Priority Sector Lending" classification.

2.3 Regulatory and Data Gaps

There is a profound "information blackout" in the sector. The GOBARdhan portal is intended to track plants, but data entry is often manual and sporadic. There is no automated, verifiable mechanism to report gas production or plant health to the Ministry of New and Renewable Energy (MNRE). This lack of transparency hampers policy formulation and makes it difficult to verify subsidy utilization or carbon abatement claims.

2.4 The Need for GFIS

Current solutions are bifurcated: expensive, imported SCADA systems for mega-plants, and zero-monitoring for rural plants. There is a critical vacuum for a cost-effective, indigenous, AI-driven solution that can democratize "process intelligence" for the Indian biogas sector. GFIS is engineered to fill this specific gap.

3. Objectives

The GreenFuel Intelligence System aims to redefine the operational standards of the Indian biogas sector. Our objectives are categorized into Technical, Economic, and Social dimensions, adhering to the SMART framework (Specific, Measurable, Achievable, Relevant, Time-bound).

3.1 Technical Objectives

1. **Develop a Robust IoT Sensor Node:** To engineer an industrial-grade embedded system (based on ESP32/STM32 architecture) capable of continuous, accurate acquisition of critical process parameters:
    - **Temperature:** ±0.5°C accuracy to monitor mesophilic/thermophilic conditions.
    - **pH & ORP:** Robust probes with auto-calibration algorithms to track acidity and redox potential.
    - **Gas Flow & Composition:** Real-time measurement of methane (% CH_4), carbon dioxide (% CO_2), and hydrogen sulfide (ppm H_2S) to ensure compliance with IS 16087:2016 standards.
2. **Implement Computer Vision for Feedstock Grading:** To create and train a Deep Learning model (utilizing MobileNetV3 or YOLOv8) that can classify incoming waste streams into "Organic," "Inorganic," and "Fibrous" categories with >95% accuracy, enabling automated feedstock quality control at the inlet.
3. **Construct a Predictive Digital Twin:** To leverage historical sensor data to train Machine Learning regressors (XGBoost/LSTM) that can predict biogas yield 24-48 hours in advance with a Mean Absolute Percentage Error (MAPE) of less than 10%, facilitating

better grid synchronization.
4. **Deploy a Cloud-Based Remote Monitoring Platform:** To establish a secure, scalable web and mobile dashboard that aggregates data from distributed plants, providing real-time alerts (via SMS/WhatsApp in vernacular languages) to operators and compliance reports to regulators.

3.2 Economic Objectives

1. **Maximize Production Efficiency:** To achieve a minimum **15% increase in biogas yield** per ton of feedstock compared to unmonitored baselines by optimizing the internal reactor environment and preventing biological inhibition.
2. **Reduce Operational Expenditure (OPEX):** To lower maintenance costs by **30%** through predictive analytics that identifies equipment faults (e.g., pump cavitation, stirrer jamming) before catastrophic failure.
3. **Enhance Revenue Realization:** To enable plant owners to maximize revenue from both CBG sales (under SATAT) and Organic Manure sales (under GOBARdhan MDA scheme) by ensuring consistent product quality and generating verifiable production logs for subsidy claims.
4. **Enable Carbon Financing:** To create a transparent, immutable data trail of methane capture that can be used to verify and sell carbon credits in voluntary carbon markets, adding a tertiary revenue stream for rural entrepreneurs.

3.3 Social and Environmental Objectives

1. **Advance the "Waste to Wealth" Mission:** To directly support the Swachh Bharat Mission (Gramin) by ensuring that community biogas plants remain functional, thereby preventing the accumulation of organic waste and the associated health hazards.
2. **Rural Skill Development & Empowerment:** To de-skill the complex operation of biogas plants through AI-driven advisory systems. This empowers local youth, women's Self-Help Groups (SHGs), and farmers to manage advanced energy assets without requiring specialized engineering degrees, fostering rural entrepreneurship.
3. **Mitigate Climate Change:** To significantly reduce fugitive methane emissions caused by inefficient digestion or gas leakages. By optimizing the capture of methane (a potent greenhouse gas), GFIS contributes directly to India's Nationally Determined Contributions (NDCs) under the Paris Agreement.

4. AI / ML Components: The Cognitive Core

The intelligence of the GFIS framework distinguishes it from standard data logging systems. It employs a multi-layered Artificial Intelligence architecture designed to mimic the decision-making capabilities of a seasoned biochemical engineer.

4.1 Computer Vision for Feedstock Quality Analysis

The quality of biogas output is deterministically linked to the quality of input. Traditional setups rely on visual inspection by laborers, which is inconsistent and subjective.

- **Technology Stack:** GFIS utilizes **Convolutional Neural Networks (CNNs)** deployed on

edge devices (Edge AI). We leverage lightweight architectures like **MobileNetV3** or **YOLOv8 (You Only Look Once)**, which are optimized for inference on low-power hardware like the Raspberry Pi 4 or NVIDIA Jetson Nano.

- **Functionality:**
  - **Inorganic Detection:** The camera mounted above the inlet hopper captures images of the waste. The model identifies contaminants such as plastics, metals, and glass. If the inorganic fraction exceeds a set threshold (e.g., 5%), the system locks the feed mechanism and triggers an alarm, preventing the "poisoning" of the digester volume.
  - **Feedstock Characterization:** The AI classifies the organic matter (e.g., leafy greens vs. woody biomass). Using pre-calibrated look-up tables for Carbon and Nitrogen content of these substrates, the system estimates the C:N ratio of the mix and advises the operator to add specific supplements (like urea or water) to balance the ratio for optimal methanogenesis.
- **Performance:** Research indicates that such models can achieve classification accuracies exceeding 93-95% in distinguishing organic from inorganic waste, significantly superior to manual sorting.

4.2 Machine Learning for Yield Prediction

Predicting gas production is crucial for supply chain logistics, especially for CBG plants that need to schedule cylinder collection trucks.

- **Algorithms:**
  - **XGBoost (Extreme Gradient Boosting):** Selected for its exceptional performance on tabular time-series data. It builds an ensemble of decision trees to model the non-linear relationship between input variables (Temp, pH, Feed Rate) and the target variable (Gas Yield). It effectively handles missing data points, which is common in rural IoT deployments.
  - **Long Short-Term Memory (LSTM):** A type of Recurrent Neural Network (RNN) capable of learning long-term dependencies. Since the retention time in a digester is 20-40 days, an operational change today (e.g., a temperature drop) impacts gas production days or weeks later. LSTM networks are uniquely suited to capture these lagged temporal correlations.
- **Training & Inference:** The models are trained on historical datasets correlating sensor telemetry with flow meter readings. The system updates its predictions every hour, providing a 24-hour and 48-hour yield forecast.

4.3 Anomaly Detection for Process Stability

Preventing biological failure is the most critical function of GFIS.

- **Unsupervised Learning:** We employ **One-Class Support Vector Machines (SVM)** or **Autoencoders** to learn the manifold of "normal" operation.
- **Mechanism:** The model continuously analyzes the correlation between parameters. For instance, in a healthy digester, pH and Temperature follow a specific stability curve. If the pH begins to drop while the Temperature remains constant, the model recognizes this as a deviation from the learned norm—a signature of VFA accumulation (early-stage acidosis).
- **Prescriptive Alerts:** Unlike a simple threshold alarm (which rings only when pH is

already low), the AI detects the *trend* toward failure. It issues a "Prescriptive Alert" advising specific actions: "Reduce feed rate by 50% and add lime buffer," effectively acting as an automated expert consultant.

5. System Architecture

The GFIS architecture is designed as a modular, scalable IoT stack, adhering to industry standards for reliability and security. It comprises four distinct layers: Perception, Network, Processing, and Application.

5.1 Layer 1: The Physical/Perception Layer

This layer consists of the hardware interface with the physical biogas plant. It is responsible for sensing the environment and executing control commands.

- **Microcontroller Unit (MCU):** The core node is built around the **ESP32** or **STM32** microcontroller. These chips are chosen for their dual-core processing capability, integrated Wi-Fi/Bluetooth, and low power consumption, making them ideal for rural energy applications.
- **Sensor Suite:**
  - **DS18B20:** Digital waterproof temperature sensors placed at the bottom, middle, and top of the digester to monitor thermal stratification.
  - **pH & ORP Probes:** Industrial-grade, gel-filled double-junction electrodes (E-201-C type) connected via signal amplification boards (like the analog pH sensor kit) to withstand the fouling nature of biogas slurry.
  - **MQ-Series / NDIR Sensors:**
    - **MQ-4:** For Methane (CH_4) detection.
    - **MG-811 / NDIR:** For Carbon Dioxide (CO_2) measurement.
    - **MQ-136:** For Hydrogen Sulfide (H_2S) detection, crucial for preventing corrosion in engines and generators.
  - **Electronic Mass Flow Meter:** To accurately measure the volume of gas produced, replacing inaccurate mechanical counters.
  - **Pressure Transducers:** To monitor dome pressure and leakages.
- **Actuators:** Relays and contactors to control stirrer motors, feed pumps, and heating elements based on logic commands from the MCU.

5.2 Layer 2: The Network/Transport Layer

Data transmission reliability is critical in rural locations where connectivity can be intermittent.

- **Edge Gateway:** A Raspberry Pi 4 acts as the local gateway. It aggregates data from the MCU sensors via Modbus/RS-485 or Wi-Fi. It performs "Edge AI" processing (running the feedstock vision model) to reduce bandwidth usage.
- **Communication Protocols:**
  - **LoRaWAN:** For large campuses or clustered community plants, Long Range (LoRa) radio is used to transmit data up to 10km to a central hub without cellular costs.
  - **4G/LTE/GSM:** For backhaul to the cloud, the gateway uses modules like SIM7600 to transmit compacted data packets via MQTT (Message Queuing Telemetry

Transport) protocol. MQTT is chosen for its lightweight overhead and ability to function on unstable networks.

5.3 Layer 3: The Processing and Cloud Layer

The "brain" of the system resides in the cloud.

- **Cloud Infrastructure:** Hosted on AWS or MeitY-empaneled Indian cloud providers to ensure data sovereignty.
- **Database Architecture:**
  - **InfluxDB:** A time-series database optimized for storing millions of timestamped sensor readings (Temperature, pH) with high write/read speeds.
  - **MongoDB:** A NoSQL database for storing unstructured data like waste images, user profiles, and maintenance logs.
- **Analytics Engine:** A Python-based microservice architecture running the Machine Learning models (XGBoost, LSTM). It processes incoming data streams, updates predictions, and identifies anomalies.

5.4 Layer 4: The Application Layer

The interface between the machine intelligence and the human operator.

- **Web Dashboard:** A React.js-based portal for plant owners and government regulators. It visualizes key performance indicators (KPIs) like Daily Yield, Methane %, and Estimated Revenue.
- **Mobile App:** A native Android application designed for field operators. It features a simplified UI (Green/Red status indicators), supports vernacular languages (Marathi, Hindi), and receives push notifications for alerts.
- **API Integration:** RESTful APIs designed to push data automatically to the **GOBARdhan Unified Registration Portal**, ensuring seamless compliance and transparency for subsidy management.

5.5 System Security

- **Device Authentication:** X.509 certificates are used to authenticate every IoT device, preventing unauthorized hardware from connecting to the network.
- **Encryption:** All data transmission is encrypted using TLS 1.2. Data at rest in the cloud is encrypted using AES-256 standards.
- **Access Control:** Role-Based Access Control (RBAC) ensures data privacy—plant owners see only their data, while aggregators or regulators see anonymized regional data.

6. Methodology

The development and deployment of GFIS follow a structured Research & Development (R&D) lifecycle, aligned with the Technology Readiness Levels (TRL) framework and the specific requirements of the DIPEX competition.

6.1 Phase I: Requirement Analysis & Lab Prototyping (Months 1-3)

- **Literature Review:** Conducted a comprehensive analysis of biogas failure modes, sensor drift issues, and existing IoT solutions.
- **Lab Setup:** Establishment of a 200-liter pilot digester at the Chatake Innoworks R&D facility in Solapur.
- **Sensor Calibration:** Testing low-cost sensors against laboratory-grade equipment. We developed software compensation algorithms to correct for pH sensor drift and temperature effects on gas sensors.

6.2 Phase II: AI Model Development (Months 3-5)

- **Data Collection:** Collaboration with local waste management centers to curate an image dataset of segregated and unsegregated waste for training the Computer Vision model.
- **Model Training:** Training the MobileNetV3 and XGBoost models using synthetic data (from biochemical simulations) and initial lab data.
- **DIPEX Preparation:** Structuring the project report and presentation according to DIPEX guidelines, focusing on the "Problem Statement," "Innovation," and "Feasibility" criteria.

6.3 Phase III: Field Trials & Pilot Deployment (Months 6-9)

- **Site Identification:** Selection of 2 operational community biogas plants in Solapur district (partnering with local Gram Panchayats or educational institutes).
- **Retrofitting:** Installation of the GFIS hardware on these live plants. The "retrofit" nature of the design is tested here—ensuring installation can be done without shutting down the plant.
- **Shadow Mode:** Running the AI system in "monitoring only" mode to validate prediction accuracy against manual logs.

6.4 Phase IV: Commercialization & Scaling (Months 9-12)

- **Product Hardening:** Designing IP65-rated enclosures to protect electronics from the corrosive, humid, and hazardous (explosive gas) environment of the plant.
- **Certification:** Initiating the process for ETDC (Electronics Test and Development Centre) testing and applying for DPIIT Startup Recognition to avail tax benefits.
- **Market Launch:** Presenting the finalized commercial prototype at DIPEX 2025 to gain industry visibility and feedback from the jury of experts.

7. Expected Outcomes

The implementation of GFIS is projected to deliver multidimensional benefits, quantifiable through Technical, Economic, and Strategic metrics.

7.1 Operational & Technical Outcomes

Metric	Baseline (Traditional Plant)	With GFIS Implementation	Improvement
Biogas Yield	Variable, often suboptimal	Optimized via pH/Temp control	**+15-20%**
Downtime	20-30 days/year (reactive)	<5 days/year (predictive)	**~80% Reduction**
Methane Content	50-55%	60-65% (via feedstock balancing)	**+10% Quality**
Process Visibility	None (Black Box)	Real-time Dashboard	**100% Visibility**

- **Predictive Maintenance:** The system is expected to prevent 90% of biological crashes by detecting acidosis trends early, saving the significant cost and time associated with re-seeding the digester.

7.2 Financial Impact & ROI Analysis

- **Revenue Augmentation:** With the government setting CBG procurement prices at remunerative levels (e.g., ~Rs 1478/mmBtu for specific periods or linked to CNG retail prices), the 15% yield increase translates directly to the bottom line.
- **Subsidy Efficiency:** By providing verifiable data, GFIS simplifies the claim process for Central Financial Assistance (CFA) under the Waste to Energy Programme (up to Rs 4 crore for larger projects) and Market Development Assistance (Rs 1500/MT) for organic manure.
- **Secondary Revenue Streams:** The verified production of high-quality Fermented Organic Manure (FOM) creates a lucrative revenue channel. FOM can be sold at Rs 3.50 - 6.00 per kg, and GFIS ensures the nutrient consistency required to command premium prices.
- **Carbon Credits:** Aggregated data from GFIS-enabled plants can be used to issue verifiable carbon credits. For a cluster of plants, this represents a significant untapped financial asset.

7.3 Strategic & Policy Impact

- **Data-Driven Policy:** Aggregated data from GFIS provides the MNRE with a high-fidelity map of the sector's performance, enabling evidence-based policy adjustments.
- **DIPEX Competitiveness:** The project's alignment with "Computational Intelligence" and "Energy" themes, coupled with its strong social impact narrative, positions it as a strong contender for awards at DIPEX 2025.
- **Energy Security:** By stabilizing decentralized biogas production, GFIS contributes to reducing India's reliance on imported LNG and crude oil, advancing the vision of *Atmanirbhar Bharat* (Self-Reliant India).

8. Long-Term Vision

Chatake Innoworks envisions GFIS not merely as a standalone product but as the foundational operating system for the "Internet of Biogas" (IoB).

8.1 Virtual Power Plants (VPP)

Our long-term goal is to aggregate thousands of decentralized GFIS-enabled biogas plants into a Virtual Power Plant. While individual rural plants are too small to interact with the national grid or large industrial buyers, a VPP of 1,000 plants represents a massive, dispatchable energy asset. GFIS will enable centralized orchestration of this asset, balancing supply and demand across the network.

8.2 Smart City Integration

As Indian cities adopt bio-methanation for municipal waste management, GFIS can integrate with Smart City Command and Control Centers (ICCC). Real-time data on digester capacity can be used to dynamically route waste collection trucks, creating a hyper-efficient "Waste Logistics" network.

8.3 Blockchain-Enabled Trust

We envision incorporating a permissioned blockchain ledger to record production data. This immutable record will serve as the "Gold Standard" for:

- **Automated Carbon Trading:** Smart contracts that automatically issue and sell carbon credits based on sensor-verified methane destruction.
- **Transparent Subsidies:** Direct Benefit Transfers (DBT) of government subsidies triggered automatically when the system verifies that the plant has met performance milestones, eliminating bureaucratic leakage.

8.4 Closing the Circular Economy Loop

GFIS will eventually close the loop between energy and agriculture. By analyzing the nutrient profile of the digestate (manure), the system will provide "Precision Agriculture" recommendations to farmers, advising them on exactly how much manure to apply for their specific crops, ensuring a truly sustainable, zero-waste ecological cycle.

9. Team & Organizational Alignment

Chatake Innoworks Pvt. Ltd. is uniquely positioned to deliver this solution, combining deep domain expertise with cutting-edge technical capabilities.

9.1 Organizational Profile

- **Division:** GreenWorks Division – Sustainable Energy Innovation Unit.
- **Location:** Neharu Industrial Estate, Solapur, Maharashtra. Our location in a major agricultural belt provides strategic access to feedstock sources (sugarcane press mud, crop residue) and a network of potential pilot sites for rapid testing and iteration.
- **Startup Status:** Registered as a Private Limited Company, we are compliant with DPIIT Startup Recognition norms (incorporated <10 years, turnover <100 Cr, innovation-focused), making us eligible for government grants and tax holidays.

9.2 Key Competencies

- **Biochemical Engineering:** In-depth understanding of anaerobic digestion dynamics, feedstock characteristics, and failure modes.
- **Embedded Systems & IoT:** In-house expertise in designing ruggedized PCBs, sensor integration, and low-power communication networks.
- **AI/ML Development:** Capability to train, deploy, and optimize complex machine learning models on edge and cloud platforms.
- **Full-Stack Development:** Expertise in building scalable, user-friendly web and mobile applications for rural users.

9.3 Project Team Structure

- **Project Lead (Principal Investigator):** Oversees system architecture, government liaison (SATAT/GOBARdhan compliance), and strategic partnerships.
- **AI & Data Scientist:** Leads the development of Computer Vision and Predictive Analytics models.
- **IoT Hardware Engineer:** Responsible for sensor selection, circuit design, and hardware ruggedization.
- **Field Operations Manager:** Manages pilot deployments, user training, and feedback loops with rural plant operators.

10. Keywords

Primary Keywords: Biogas Optimization, Artificial Intelligence (AI), Internet of Things (IoT), Sustainable Energy, SATAT Scheme, GOBARdhan Initiative, Anaerobic Digestion, Predictive Maintenance, Machine Learning, Smart Infrastructure.
Secondary Keywords: Compressed Biogas (CBG), Green Energy, Circular Economy, DIPEX 2025, DPIIT Startup, Waste to Wealth, Renewable Energy Intelligence, Embedded Systems, Computer Vision, Carbon Credits, Rural Development, Fermented Organic Manure (FOM), Virtual Power Plant (VPP), Edge Computing, Methane Abatement.
Technical Nomenclature:

- **CNN (Convolutional Neural Network):** Deep learning algorithm used for image analysis (feedstock grading).
- **XGBoost:** Extreme Gradient Boosting, a machine learning algorithm used for regression (yield prediction).
- **LSTM (Long Short-Term Memory):** Recurrent Neural Network architecture for time-series forecasting.
- **C:N Ratio:** Carbon-to-Nitrogen ratio, a critical parameter for microbial health in digesters.
- **HRT (Hydraulic Retention Time):** The average time organic material remains in the digester.
- **VFA (Volatile Fatty Acids):** Intermediate compounds in digestion; accumulation indicates instability.

Innoworks Pvt. Ltd. Address: Neharu Industrial Estate, Solapur, Maharashtra, India.

Works cited

1. HPCL SATAT Initiative - Hindustan Petroleum, https://www.hindustanpetroleum.com/pages/satat 2. SATAT: A Sustainable Step Toward Energy Independence - raghavdas.in, https://raghavdas.in/satat-a-sustainable-step-toward-energy-independence/ 3. Sustainable Alternative Towards Affordable Transportation - Vikaspedia - Energy, https://energy.vikaspedia.in/viewcontent/energy/policy-support/renewable-energy-1/biofuels/sustainable-alternative-towards-affordable-transportation?lgn=en 4. Methane Mitigation: Lessons Learned from India's Compressed Biogas (Renewable Natural Gas) Projects January 2025, https://www.globalmethane.org/documents/CBG_Report_%20January%202025.pdf 5. galvanizing organic bio-agro resources dhan under swachh bharat mission (gramin) april 2018, http://swachhbharatmission.gov.in/SLRM/Download.aspx?FileName=Final%20GOBAR-Dhan%20Policy%20Document.pdf 6. Petroleum Minister launches SATAT initiative to promote Compressed Bio-Gas as an alternative, green transport fuel, https://www.pib.gov.in/PressReleasePage.aspx?PRID=1548031 7. Market Report - World Biogas Association, https://www.worldbiogasassociation.org/wp-content/uploads/2023/12/FINAL_COUNTRY_REPORT.with-authors.pdf 8. With no transparency on CBG production data, purpose of GOBARdhan portal remains unserved - Down To Earth, https://www.downtoearth.org.in/energy/with-no-transparency-on-cbg-production-data-purpose-of-gobardhan-portal-remains-unserved 9. A Low-Cost IoT System Based on the ESP32 Microcontroller for Efficient Monitoring of a Pilot Anaerobic Biogas Reactor - MDPI, https://www.mdpi.com/2076-3417/15/1/34 10. Smart Biogas Plant Monitoring System Using Internet of Things (IoT) Technology - SciTePress, https://www.scitepress.org/Papers/2022/119556/119556.pdf 11. Waste Classification for Sustainable Development Using Image Recognition with Deep Learning Neural Network Models - MDPI, https://www.mdpi.com/2071-1050/14/12/7222 12. Classification Organic and Inorganic Waste with Convolutional Neural Network Using Deep Learning | International Journal of Intelligent Systems and Applications in Engineering, https://ijisae.org/index.php/IJISAE/article/view/2637 13. Biogas Production Prediction Based on Feature Selection and Ensemble Learning - MDPI, https://www.mdpi.com/2076-3417/14/2/901 14. Machine learning methods for the modelling and optimisation of biogas production from anaerobic digestion: a review - PubMed, https://pubmed.ncbi.nlm.nih.gov/38376778/ 15. Targets under SATAT Scheme, https://www.pib.gov.in/PressReleasePage.aspx?PRID=1881749 16. IREDA Loan Scheme for “Compressed Bio- Gas (CBG)” under SATAT Scheme, https://www.ireda.in/images/HTMLfiles/20_Compressed%20Biogas%20SATAT.pdf 17. Market Development Assistance Guidelines - GOBARdhan Unified Registration Portal, https://gobardhan.eil.co.in/whats-new/Market-Development-Assistance-Guidelines.pdf 18. DPIIT Startup Recognition & Tax Exemption, https://www.startupindia.gov.in/content/sih/en/startupgov/startup_recognition_page.html 19. Registration as a Startup - NSWS, https://www.nsws.gov.in/portal/approval-details/ministry-of-commerce-and-industry/dpiit/registration-as-a-startup 20. dipex – 2025 - Directorate of Technical Education, Maharashtra State, India, https://dte.maharashtra.gov.in/wp-content/uploads/2025/01/DIPEX-2025-DTE-Letter.pdf 21. Waste to Energy Programme,

https://gobardhan.sbm.gov.in/whats-new/Waste_to_Energy_Program%20revised%20guidelines.pdf 22. Biogas Programme (Phase-I) for FY 2021-22 to 2025-26, https://biogas.mnre.gov.in/about-the-programmes 23. Guidelines for setting up of Biogas plants under GOBARdhan scheme of SBM-G - Panchayati Raj & Drinking Water Department - Government Of Odisha, https://panchayat.odisha.gov.in/sites/default/files/2023-09/Guideline%20for%20setting%20up%20of%20Biogas%20plant%20under%20GOBARdhan%20Scheme%20of%20SBM-G.pdf 24. The Selection of Biogas Plants in the Indian Context Based on Performability—An Analytic Hierarchy Process and Weighted Aggregated Sum Product Assessment Approach - MDPI, https://www.mdpi.com/2673-3994/5/2/13 25. INTELLIGENT ORGANIC RECYCLABLE OBJECTS CLASSIFICATION SYSTEM USING MACHINE LEARNING FOR LANDFILL MINIMIZATION - Dialnet, https://dialnet.unirioja.es/descarga/articulo/9906283.pdf 26. Rs 1,478/mmBtu: Centre raises CBG procurement price - Iamrenew.com, https://www.iamrenew.com/policy/rs-1478-mmbtu-centre-raises-cbg-procurement-price-to-boost-green-fuel/ 27. India's CBG sector gains momentum, but challenges remain - REGlobal, https://reglobal.org/indian-cbg-sector-gains-momentum-but-challenges-remain/ 28. Development of an IoT System for Monitoring the Operation of a Biogas Plant - Preprints.org, https://www.preprints.org/manuscript/202507.0368 29. Development of an IoT System for Monitoring the Operation of a Biogas Plant - Sciety, https://sciety.org/articles/activity/10.20944/preprints202507.0368.v1 30. Galvanizing Organic Bio-Agro Resources Dhan (GOBARdhan) - myScheme, https://www.myscheme.gov.in/schemes/gobardhan 31. Unnat Bharat Abhiyan (UBA) - Rural Development & Innovation | Central University of Gujarat, https://www.cug.ac.in/uba.php 32. SATAT (Sustainable Alternative Towards Affordable Transportation) - Indian Oil, https://iocl.com/pages/satat-overview 33. Artificial Intelligence Based System for Sorting and Detection of Organic and Inorganic Waste, https://thesai.org/Downloads/Volume16No5/Paper_9-Artificial_Intelligence_Based_System.pdf 34. A Review of Machine Learning Models in Predicting Biogas Production - AIP Publishing, https://pubs.aip.org/aip/acp/article-pdf/doi/10.1063/5.0181016/19716021/060006_1_5.0181016.pdf 35. Comparison of different machine learning algorithms to estimate liquid level for bioreactor management - Environmental Engineering Research, https://www.eeer.org/journal/view.php?number=1396 36. Design and Implementation of IoT-Enabled Device for Real-Time Monitoring of Greenhouse Gas Emissions, and Pressure in Anaerobic Reactors - IEEE Xplore, https://ieeexplore.ieee.org/iel8/6287639/10380310/10643995.pdf 37. SMART IOT BASED BIOGAS MONITORING SYTEM - ijrpr, https://ijrpr.com/uploads/V6ISSUE2/IJRPR38932.pdf 38. IoT Architecture Explained: 6 Layers, Components, & Diagrams - Minew, https://www.minew.com/iot-architecture/ 39. An IoT based Condition Monitoring System of Biogas Electrical Generator for Performance Evaluation - Semantic Scholar, https://pdfs.semanticscholar.org/1935/82a60e77ff4a420e1cd22e94200e9d2d4173.pdf 40. An Illustrative Example of an IoT System Architecture Composed of... | Download Scientific Diagram - ResearchGate, https://www.researchgate.net/figure/An-Illustrative-Example-of-an-IoT-System-Architecture-Composed-of-Heterogeneous-Data_fig5_341257704 41. What Is IoT Architecture? | MongoDB, https://www.mongodb.com/resources/basics/cloud-explained/iot-architecture 42. DIPEX 2025 Idea Presentation Guidelines | PDF - Scribd, https://www.scribd.com/document/827621563/Idea-presentation-round-guidelines-for-participants 43. Dipex profile1 | PDF, https://www.slideshare.net/slideshow/dipex-profile1/15815282 44.

Organic Fertilizers and Manure - animal based fertilizer Latest Price, Manufacturers & Suppliers - IndiaMART, https://dir.indiamart.com/impcat/organic-fertilizers.html 45. Organic Manure at Best Price in New Delhi, Delhi | Ecological Products Industries - Tradeindia, https://www.tradeindia.com/products/organic-manure-c892132.html 46. Visualizing IoT architecture with diagrams - Kaa IoT platform, https://www.kaaiot.com/iot-knowledge-base/visualizing-iot-architecture-with-diagrams 47. Recognition Page - Startup India, https://www.startupindia.gov.in/content/sih/en/recognition-page.html